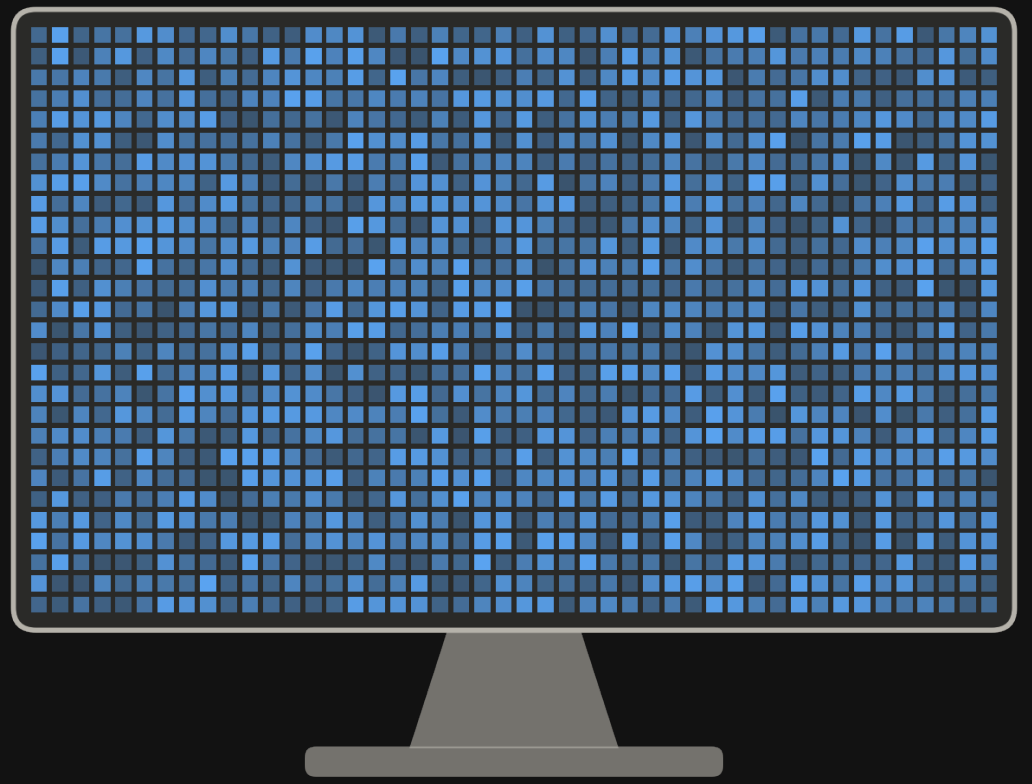
Saving the Tokens

Your agent pays again for everything it already sent.

Find the big thing. Stop sending it.



Shyamji Tiwari

Applied Scientist, Amazon

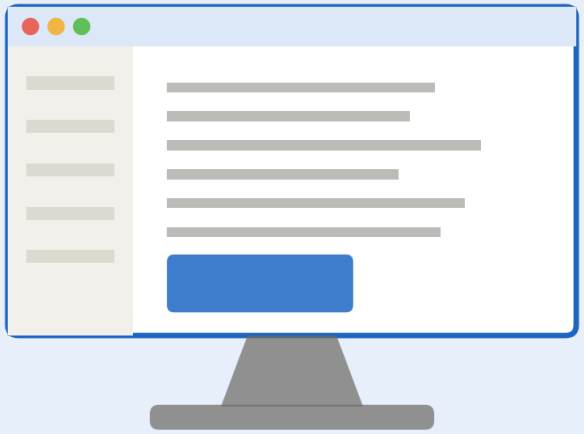
IEEE AgenticAI Summit 2026 · Bangalore

Views are my own.

HANDS UP

Which costs more to send to your AI agent?

A



One screenshot

of your screen

or

B



The system prompt

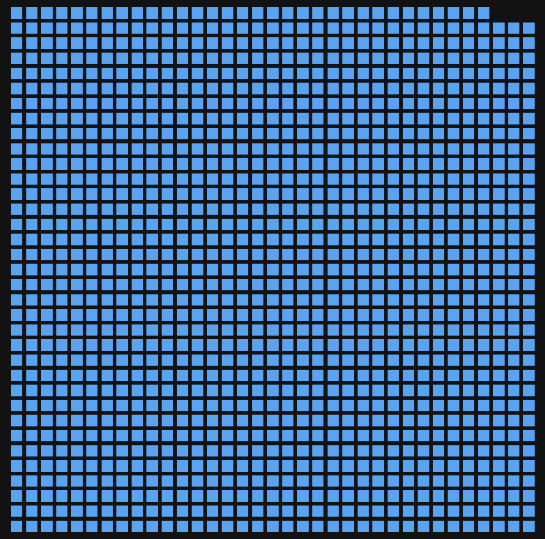
instructions sent every message (a short one)

Agent = an AI that reads your screen and acts. **Turn** = one message to it, and its reply.

3.1×

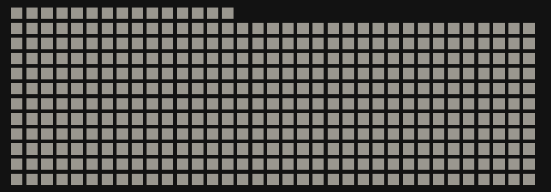
One screenshot is three short prompts.

And it sends a new one **every turn.**



1,222

one 1920×1080 screenshot, 1,280-token cap



400

a short system prompt

1,280-token image cap · 400 tokens is a short prompt; real ones often run 2,000+

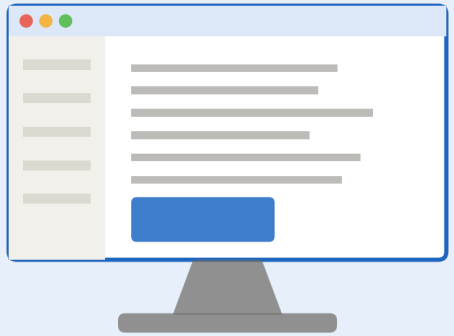
A token is the unit of the bill

You pay for everything going **in**, not just what comes out.



750

tokens · a dense page of text



1,222

tokens · one capped screenshot



200

tokens · 8 seconds of speech

screenshot at a 1,280-token cap · speech at 25 tokens per second

Every call sends the whole history again

Pseudo-code of almost every agent. The usage field is where the bill is read.

```
history = [system_prompt]
while not done:
    history.append(observe())   # screenshot, speech, tool output
    reply = model(history)      # billed for ALL of history
    log(reply.usage)            # input/output tokens = the bill
    history.append(reply)
```

Orange = what you pay for again. **Blue** = where you read what it cost.

How a picture becomes tokens

Using the “cut into tiles” strategy (slide 3 used a cap; next slide compares them)

STEP 1

Your screenshot



1920×1080 pixels

STEP 2

Cut into tiles



a 4×2 grid of squares

STEP 3

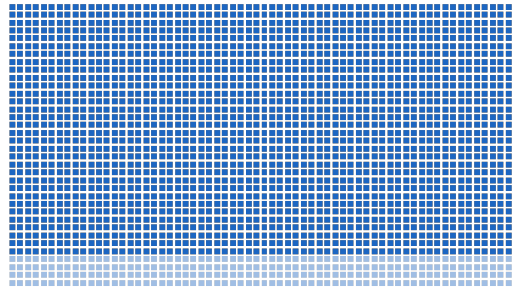
256 tokens per tile



8 tiles, plus 1 small copy of the whole image

STEP 4

2,304 tokens



one square = one token

9 tiles × 256 tokens = **2,304 tokens**

Vision models mostly use one of three strategies

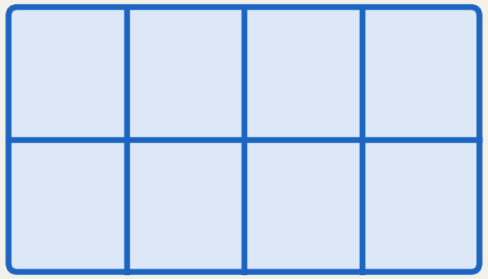
All three priced on the same 1920×1080 screenshot



Squash it

Same cost, any size

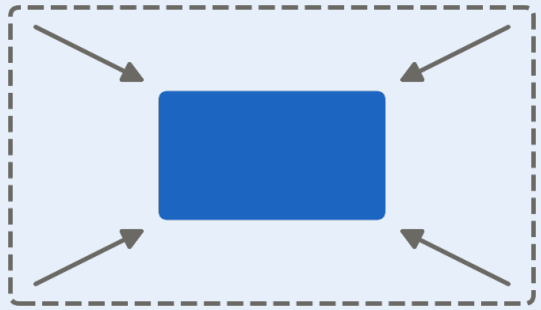
576
tokens



Cut into tiles

Cost follows the shape

2,304
tokens



Shrink to a budget

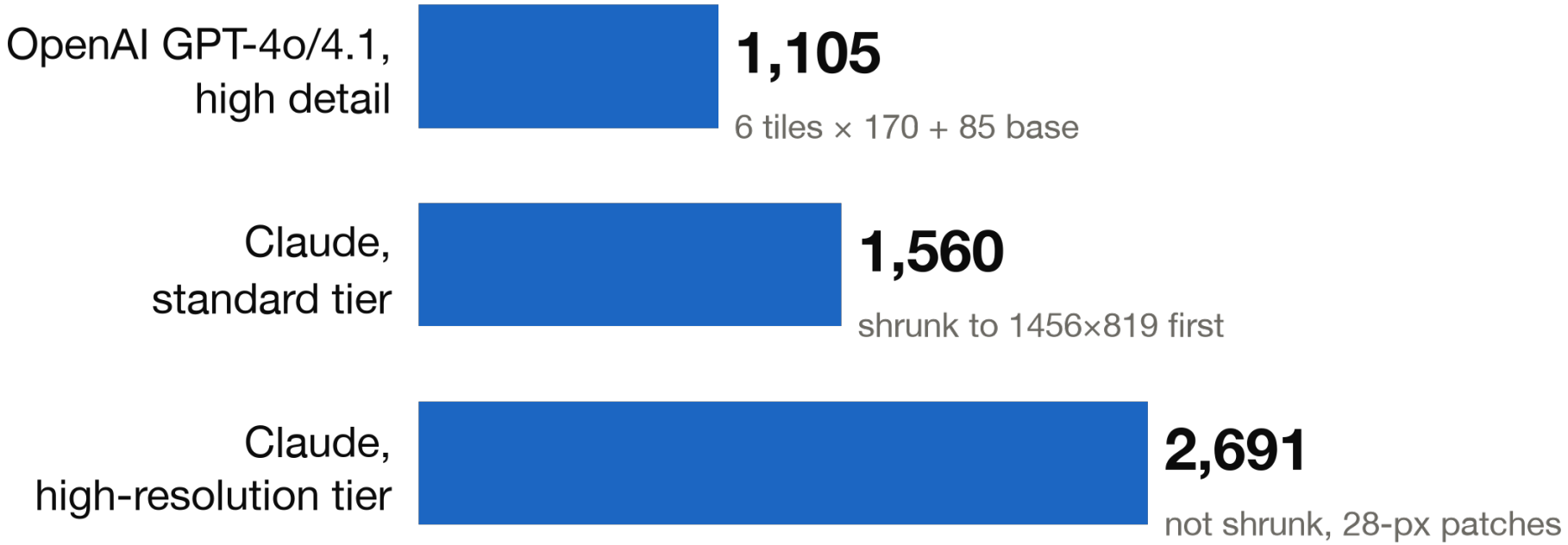
You set the cap

1,222
tokens

the rest of this talk uses “shrink to a budget”, capped at 1,280

Real products price it differently

A 1920×1080 screenshot, priced by each vendor's published rule



Claude standard: a 4K screen costs the same **1,560**. The cap sets the bill.

Claude 4.7+ is the high-res tier: an upgrade can move it from 1,560 to 2,691.

published rules, checked 2026-10-29 · vendors change them, re-check before quoting

Open a terminal in the folder, run this

Look for **PASS**

No installs. No API key.
No internet needed.

```
$ python3 check_setup.py

7 checks ... ok
PASS. You are ready.
```

Get the files

github.com/ieee-ai/agentic-ai-26

folder Saving-the-Tokens

Windows: use py, not python3

No Python? Open calc.html

How to get a terminal in the folder

1 Mac: ⌘-Space, type Terminal. Windows: Start, type PowerShell.

2 Type cd and a space, drag the folder into the window, press Enter.

EXERCISE 1

What does one image cost?

```
$ python3 ex1_image.py --size 1920x1080
modelled strategies
Squash to a square      576
Tiles (up to 12)      2,304
Tiles (up to 6)         768
Budget 1,280          1,222
Budget 256              252
real products (published rules)
Claude standard       1,560
Claude hi-res         2,691
OpenAI GPT-4o         1,105
```

Same picture: **252** to **2,691** tokens, depending on the strategy.

TRY IT

1. Your own screen size
   Mac: menu > About This Mac. Windows: Settings > Display.
2. Which is cheapest?
3. Double the width
4. Try 3840x2160

Four times the pixels. Same bill.

Using the “cut into tiles” strategy

1280×800

1.0 million pixels

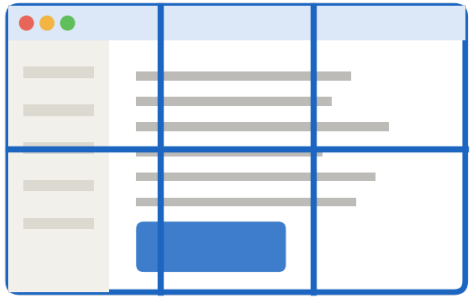


1,792 tokens

3×2 grid

1920×1200

2.3 million pixels

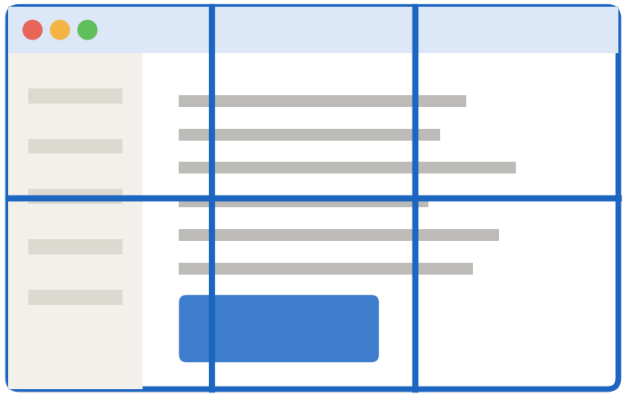


1,792 tokens

3×2 grid

2560×1600

4.1 million pixels



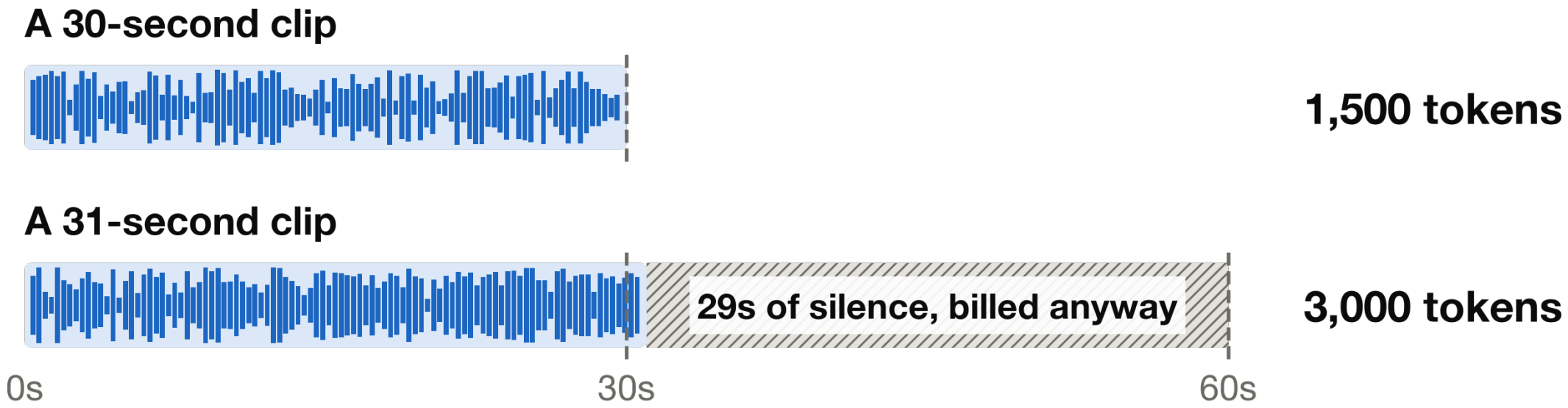
1,792 tokens

3×2 grid

Tiling looks at the shape of your image, not its size.

With audio, you can pay for silence

Some audio models pad every clip to a 30-second window



33 tokens

Send the words, not the sound.

An 8-second sentence as text: 6× cheaper than 25 tokens/s audio, 45× cheaper than a padded clip.

Every turn re-sends the whole conversation

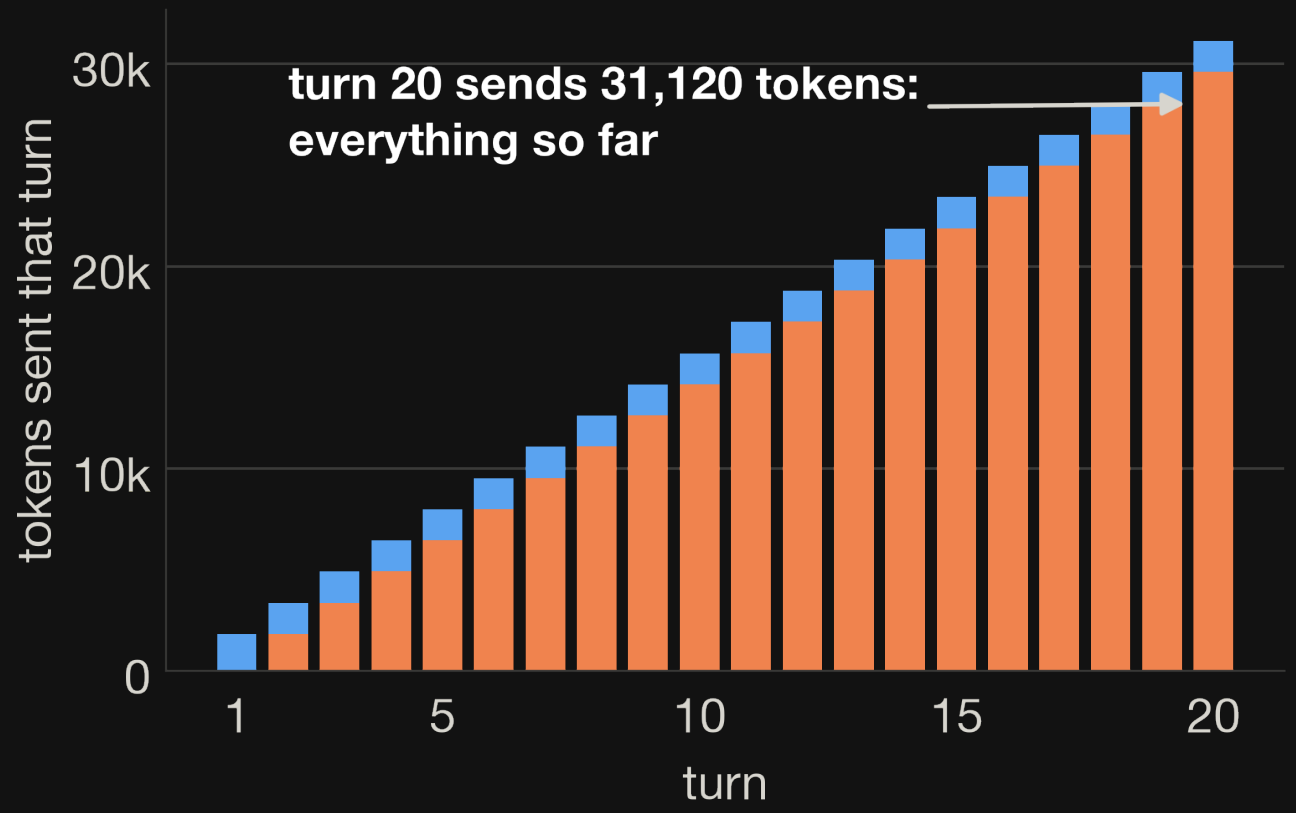
91%

of the bill was sent before
(20 turns, no caching)

New content **31,120**

Sent again **298,300**

Billed in total **329,420**

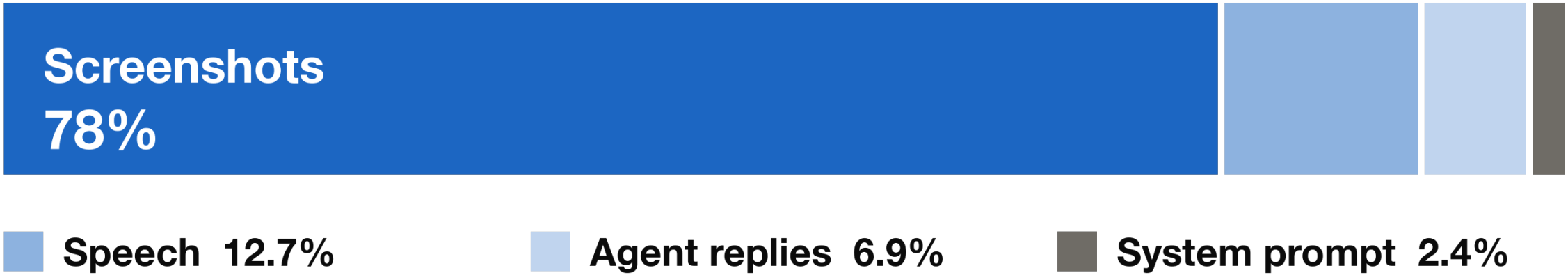


Simulated 20-turn session · illustrative settings · input tokens only

Your bill is a multiplication

how **big** each thing is × how many times you **send it again**

Share of the bill, one 20-turn session



About **$0.99** a session, **$296,478** a month at 10,000 sessions a day. Assumed $3.00 per million tokens.

Simulated 20-turn session · illustrative settings · input tokens only

EXERCISE 3

Ten ways to cut the bill. Which wins?

For a voice agent that watches the screen, 20 turns

1. Cut the system prompt by 60%
2. Drop old screenshots, keep the newest
3. Halve every agent reply
4. Halve the tokens per second of speech
5. Keep only the last 5 turns
6. Pay 10% for repeated text (prefix caching)
7. Screenshot every 4th turn
8. Send a transcript, not the audio
9. Send the screen's text, not a picture (~300 tokens)
10. Shrink the image budget

ON PAPER, NOW

Most?

Least?

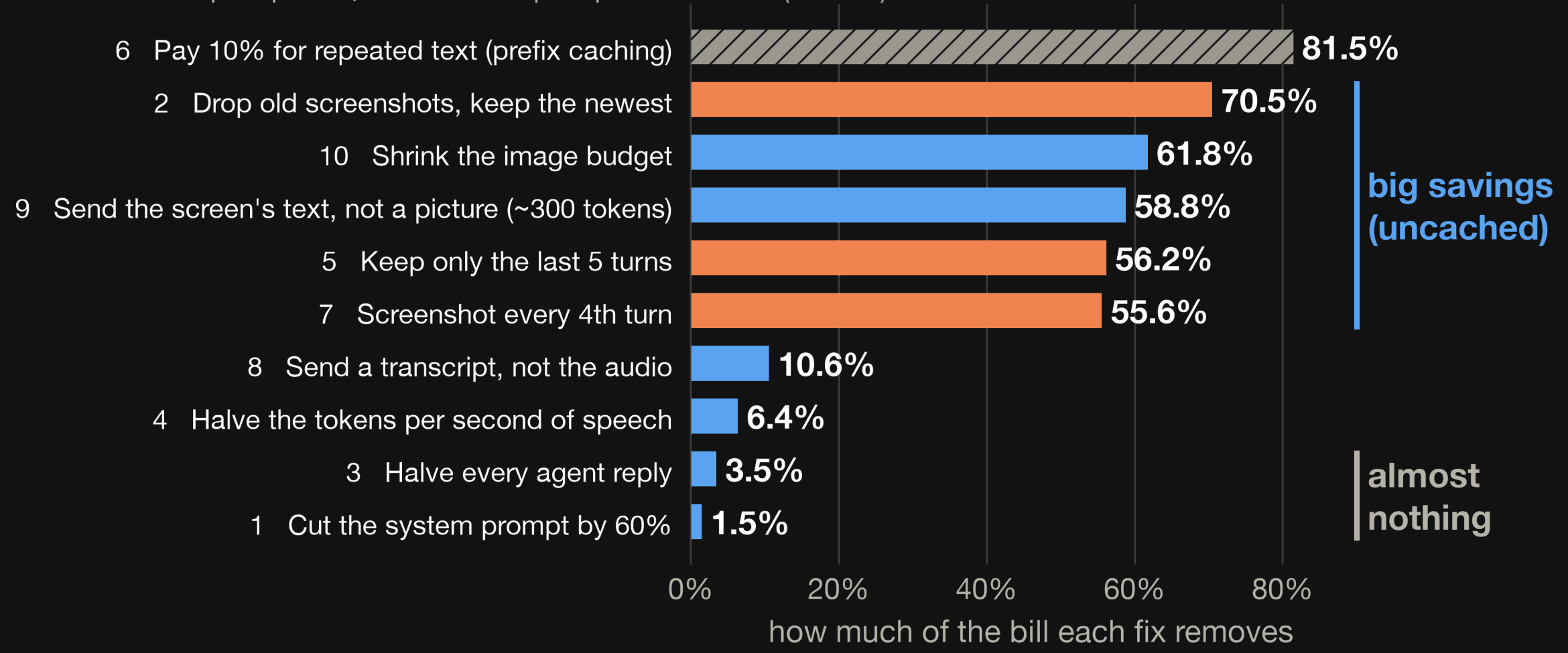
90 seconds. Then run:

```
python3 ex3_budget.py
```

EXERCISE 3 · THE ANSWER

Ranked by what it actually saves

With a 400-token prompt. At 4,000 tokens the prompt fix saves 12% (slide 18).

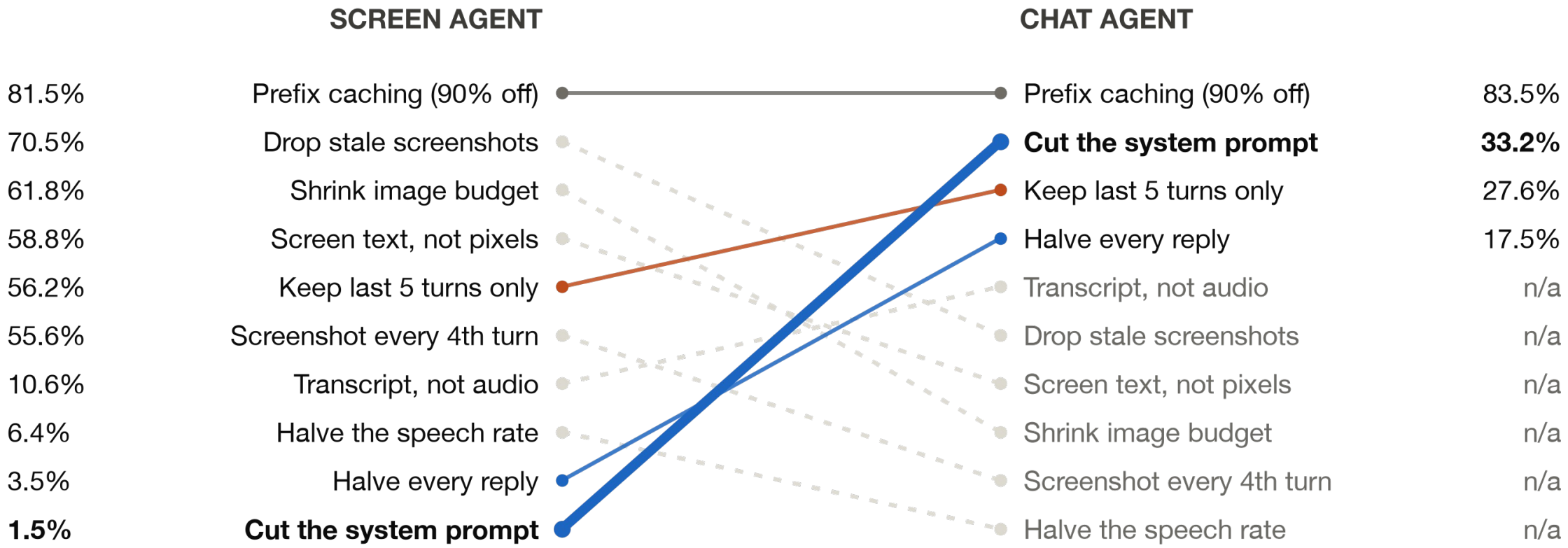


Make it smaller · Send it again less · Not a design change

best 70% · worst 1.5%

Run it on a different agent. The ranking flips.

Same ten fixes. A chat agent: 6,000-token system prompt, no pictures



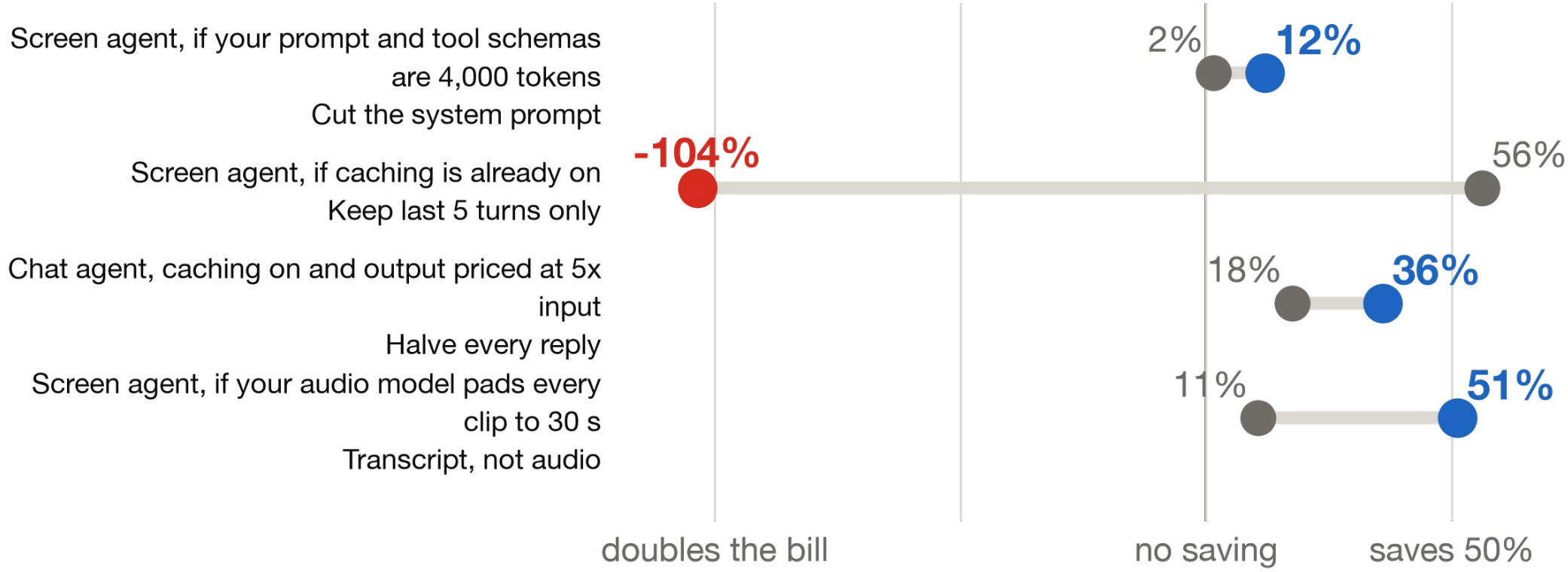
Prompt cut: **last** at 1.5%, then **#2** at 33.2%

```
python3 ex3_budget.py --agent both
```


Simulated 20-turn session · illustrative settings · input tokens only

That ranking is not a law

Change one assumption of the agent and a fix moves a long way



Measure your own setup. These settings are illustrative, not any one model's.

TAKE THIS WITH YOU

Monday morning: find YOUR big number

1

Log input tokens on every model call.

2

Run `bill_report.py` on the log.

3

See what dominates: the prefix, or what each turn adds.

4

Pick the fix aimed at that one.

Your bill = how big each item is × how often you re-send it

Dropping old screenshots saved **70%** (38% with caching)

Cutting the prompt: **1.5%** vs **33%** on a chat agent

Thank you

Shyamji Tiwari
Applied Scientist, Amazon

github.com/ieee-ai/agentic-ai-26/Saving-the-Tokens

Simulated 20-turn session · illustrative settings · input tokens only